EXPERIMNET 3

1. **AIM:**

   To create a Star Schema data model in Microsoft Power BI using the Global Superstore data.

2. **OBJECTIVES:**

   - To understand the structure of a Star Schema.
   - To identify suitable fact and dimension attributes from business data.
   - To organize transactional data into a dimensional model.
   - To create separate dimension tables for analysis.
   - To connect dimension tables with a central fact table.
   - To use DAX measures for analyzing business performance.
   - To verify the accuracy of the developed data model.

   - **TOOLS / SOFTWARE REQUIRED:**
   - Microsoft Power BI Desktop
   - Global Superstore Dataset.xlsx
   - Power Query Editor – built into Power BI Desktop
   - DAX (Data Analysis Expressions)

3. **DATASET USED:** Global Superstore Dataset.xlsx.

4. **THEORY:**

   Dimensional Modeling is a method of organizing data into structures that make business analysis and reporting easier. It mainly consists of **Fact Tables** and **Dimension Tables**.

   Fact Table stores business transactions and measurable values such as Sales, Profit, Quantity, and Discount. It generally forms the center of the analytical model.

   Dimension Tables contain descriptive information used to analyze the facts. Examples include Customer, Product, Region, and Date.

   Star Schema is a dimensional model where a central fact table is directly connected to multiple dimension tables. This structure makes the data model simple, organized, and easier to use for reporting.

   In this experiment, the Global Superstore **Orders** data is organized into a central Fact_Sales table. Separate tables such as Dim_Customer, Dim_Product, Dim_Region, and Dim_Date provide descriptive information for analysis.

5. PROCEDURE:

Part A: Load Dataset

Step 1: Open Microsoft Power BI Desktop.

Launch Microsoft Power BI Desktop.

Step 2: Import the Global Superstore dataset.

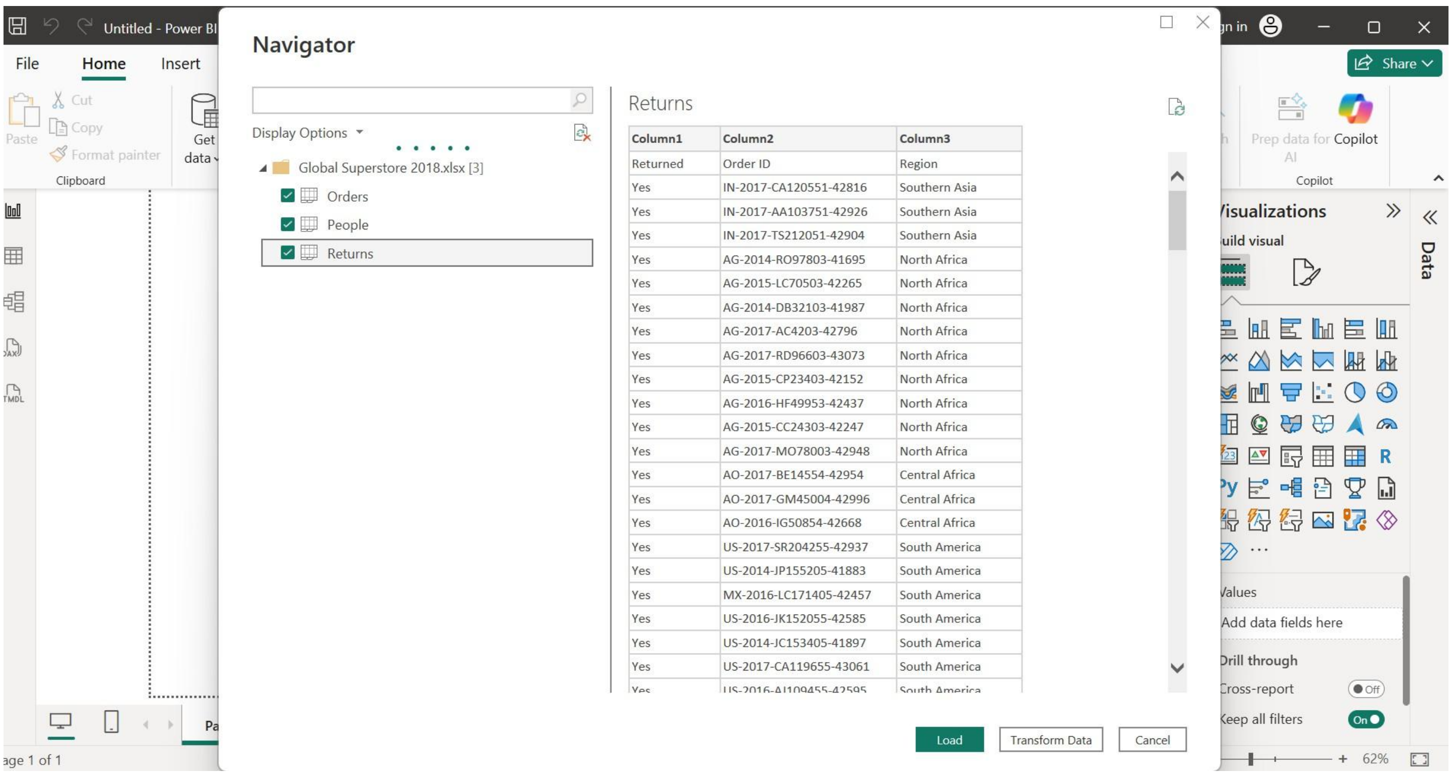
Home > Get Data > Excel Workbook

Select: Global Superstore Dataset.xlsx

Click Load.

Step 3: Select the required worksheets.

Import the following worksheets:



- Orders
- Returns
- People

Verify that the imported tables appear in Power BI.

Part B: Identify Fact and Dimension Tables

Step 4: Analyse the Orders table.

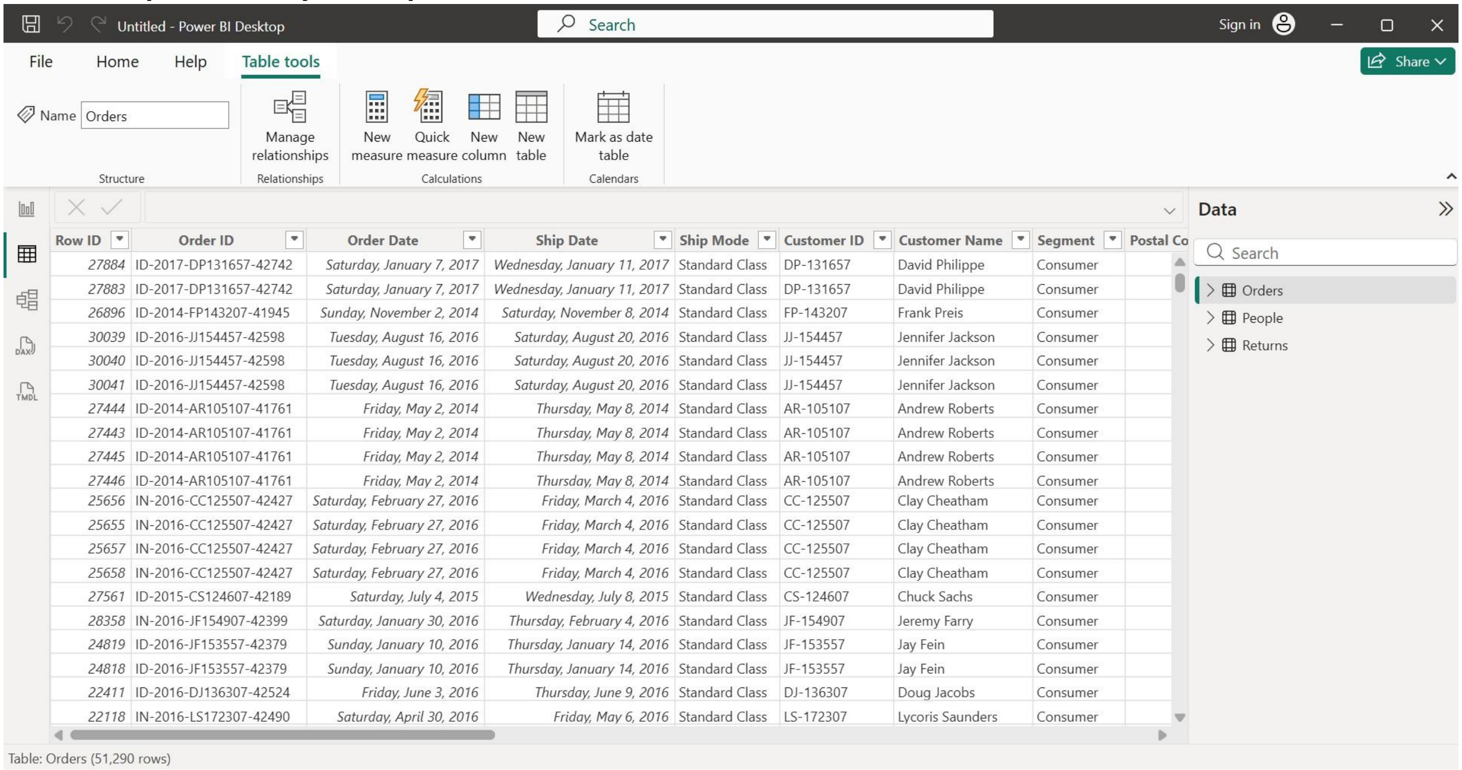
Examine the Orders table and identify measurable business attributes such as:

- Sales
- Profit
- Quantity

- Discount

These attributes represent measurable business events and are suitable for the Fact Table.

Step 5: Identify descriptive attributes.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Co
27884	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
27883	ID-2017-DP131657-42742	Saturday, January 7, 2017	Wednesday, January 11, 2017	Standard Class	DP-131657	David Philippe	Consumer	
26896	ID-2014-FP143207-41945	Sunday, November 2, 2014	Saturday, November 8, 2014	Standard Class	FP-143207	Frank Preis	Consumer	
30039	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30040	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
30041	ID-2016-JJ154457-42598	Tuesday, August 16, 2016	Saturday, August 20, 2016	Standard Class	JJ-154457	Jennifer Jackson	Consumer	
27444	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27443	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27445	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
27446	ID-2014-AR105107-41761	Friday, May 2, 2014	Thursday, May 8, 2014	Standard Class	AR-105107	Andrew Roberts	Consumer	
25656	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25655	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25657	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
25658	IN-2016-CC125507-42427	Saturday, February 27, 2016	Friday, March 4, 2016	Standard Class	CC-125507	Clay Cheatham	Consumer	
27561	ID-2015-CS124607-42189	Saturday, July 4, 2015	Wednesday, July 8, 2015	Standard Class	CS-124607	Chuck Sachs	Consumer	
28358	IN-2016-JF154907-42399	Saturday, January 30, 2016	Thursday, February 4, 2016	Standard Class	JF-154907	Jeremy Farry	Consumer	
24819	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
24818	ID-2016-JF153557-42379	Sunday, January 10, 2016	Thursday, January 14, 2016	Standard Class	JF-153557	Jay Fein	Consumer	
22411	ID-2016-DJ136307-42524	Friday, June 3, 2016	Thursday, June 9, 2016	Standard Class	DJ-136307	Doug Jacobs	Consumer	
22118	IN-2016-LS172307-42490	Saturday, April 30, 2016	Friday, May 6, 2016	Standard Class	LS-172307	Lycoris Saunders	Consumer	

Identify descriptive attributes related to:

- Customer
- Product
- Region
- Date

These attributes are used to create Dimension Tables.

Step 6: Categorize the data.

Organize the attributes into:

A. Fact Table: Fact_Sales

B. Dimension Tables: Dim_Customer, Dim_Product, Dim_Region, Dim_Date

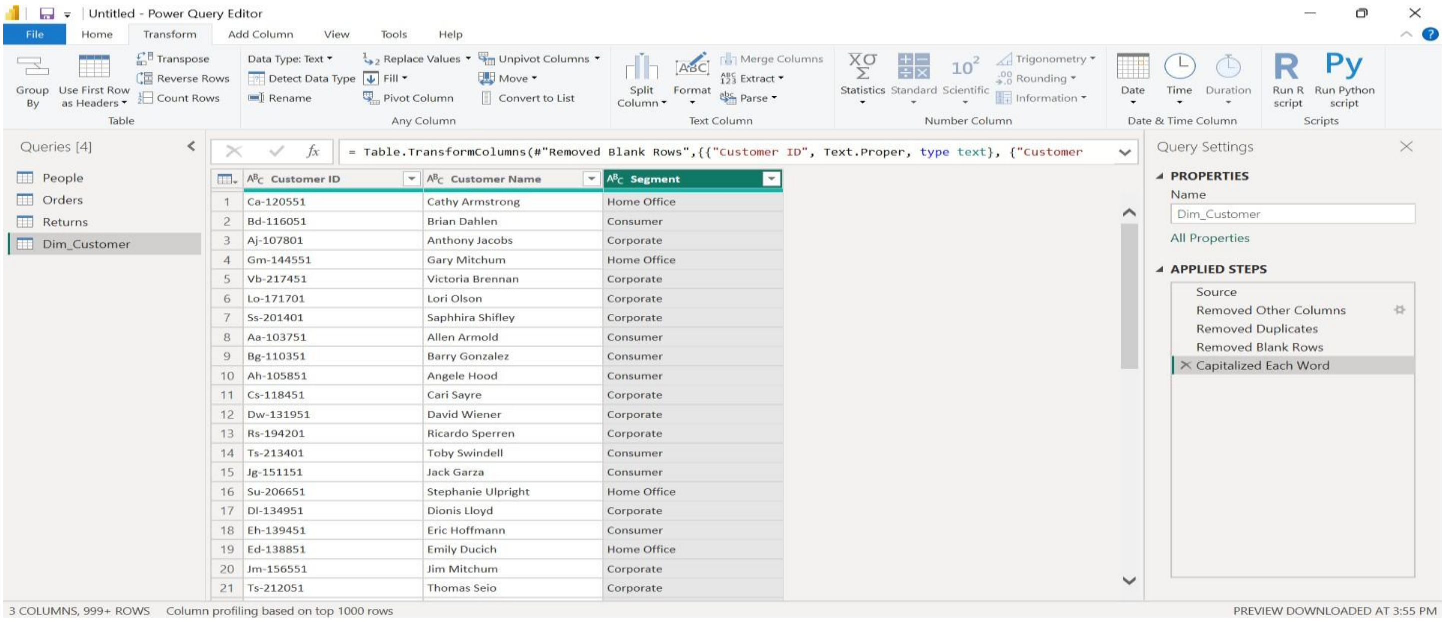
This classification forms the basis of the Star Schema.

Part C: Create Dimension Tables

Step 7: Open Power Query Editor.

Home > Transform Data

Step 8: Create Dim_Customer.

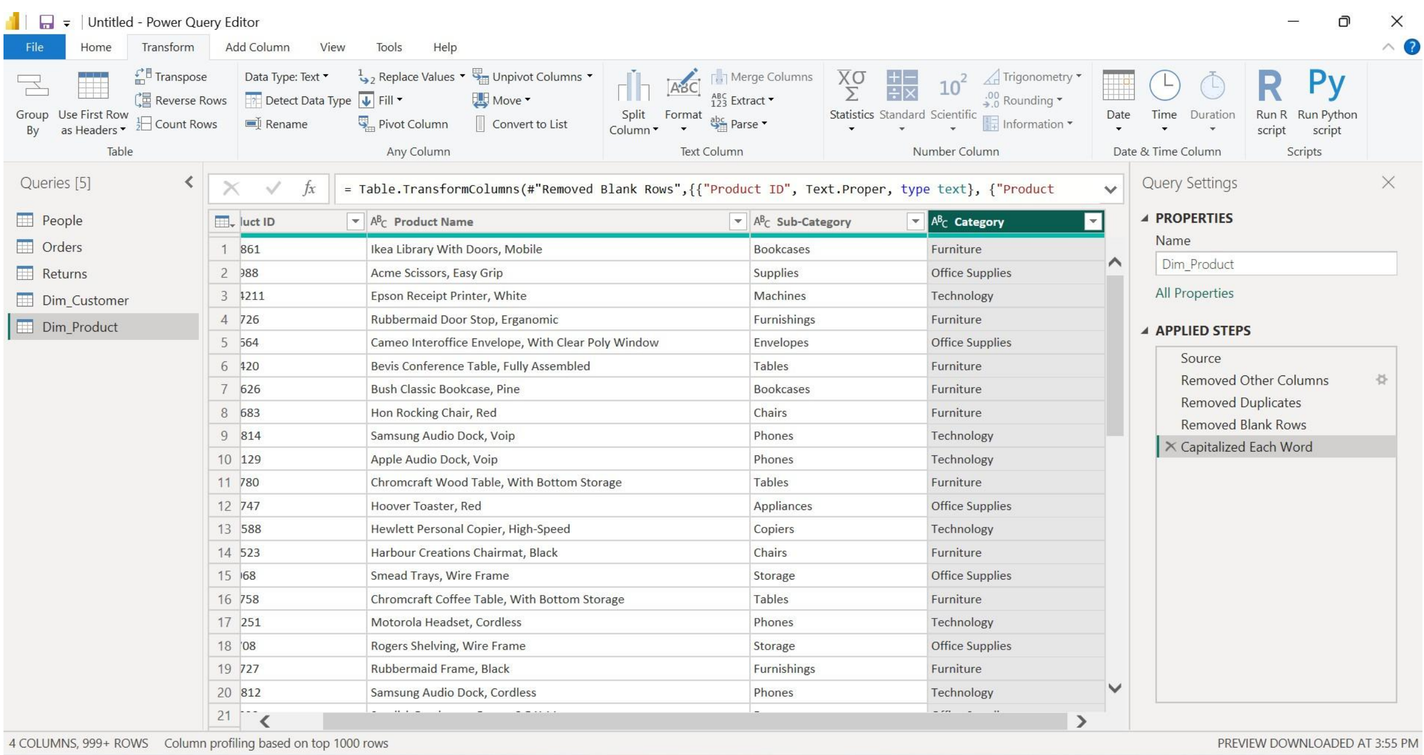


Create a customer dimension containing:

- Customer ID
- Customer Name
- Segment

Remove duplicate records from the dimension.

Step 9: Create Dim_Product.



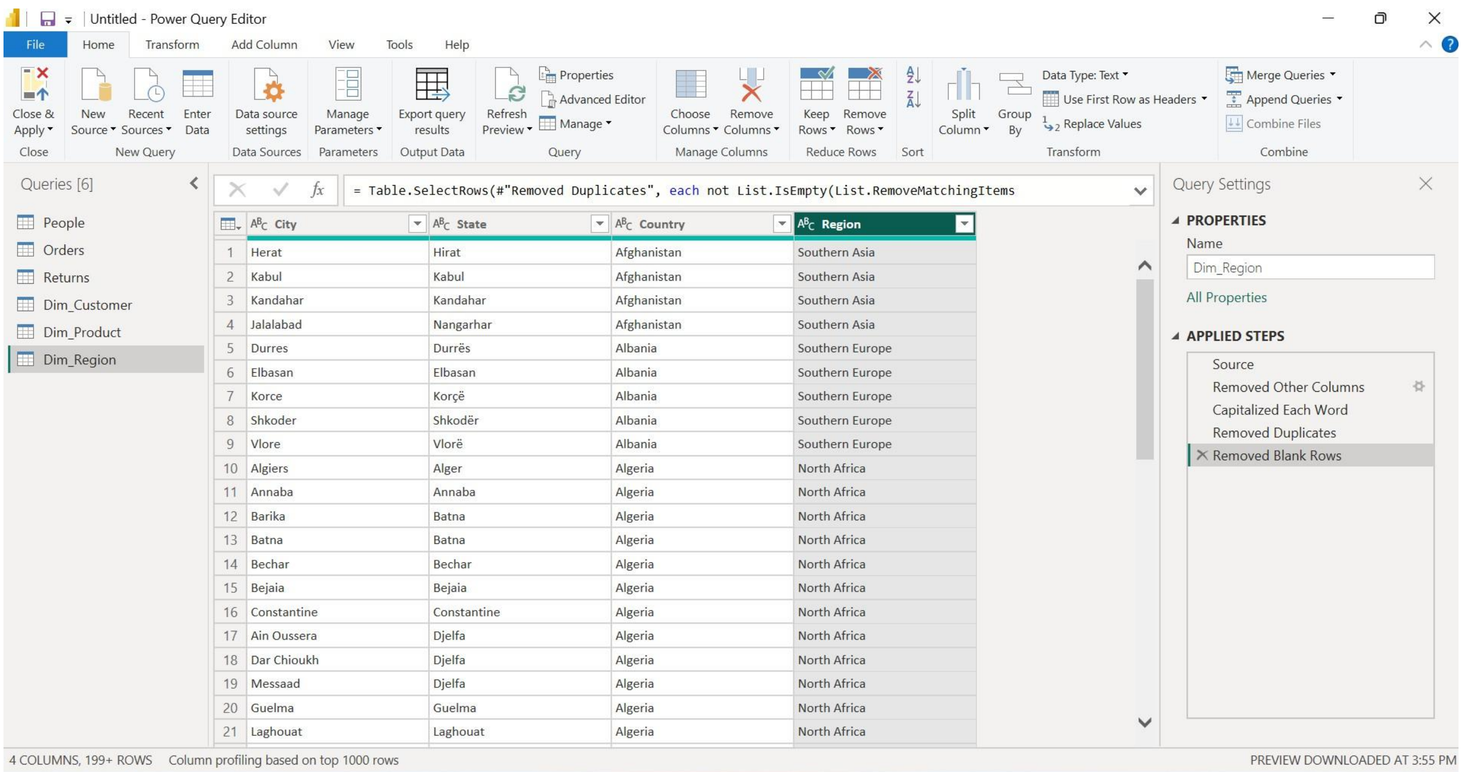
Create a product dimension containing:

- Product ID
- Product Name
- Category
- Sub-Category

Remove duplicate records.

Step 10: Create Dim_Region.

Create a region dimension containing:



- Region
- Country
- State
- City

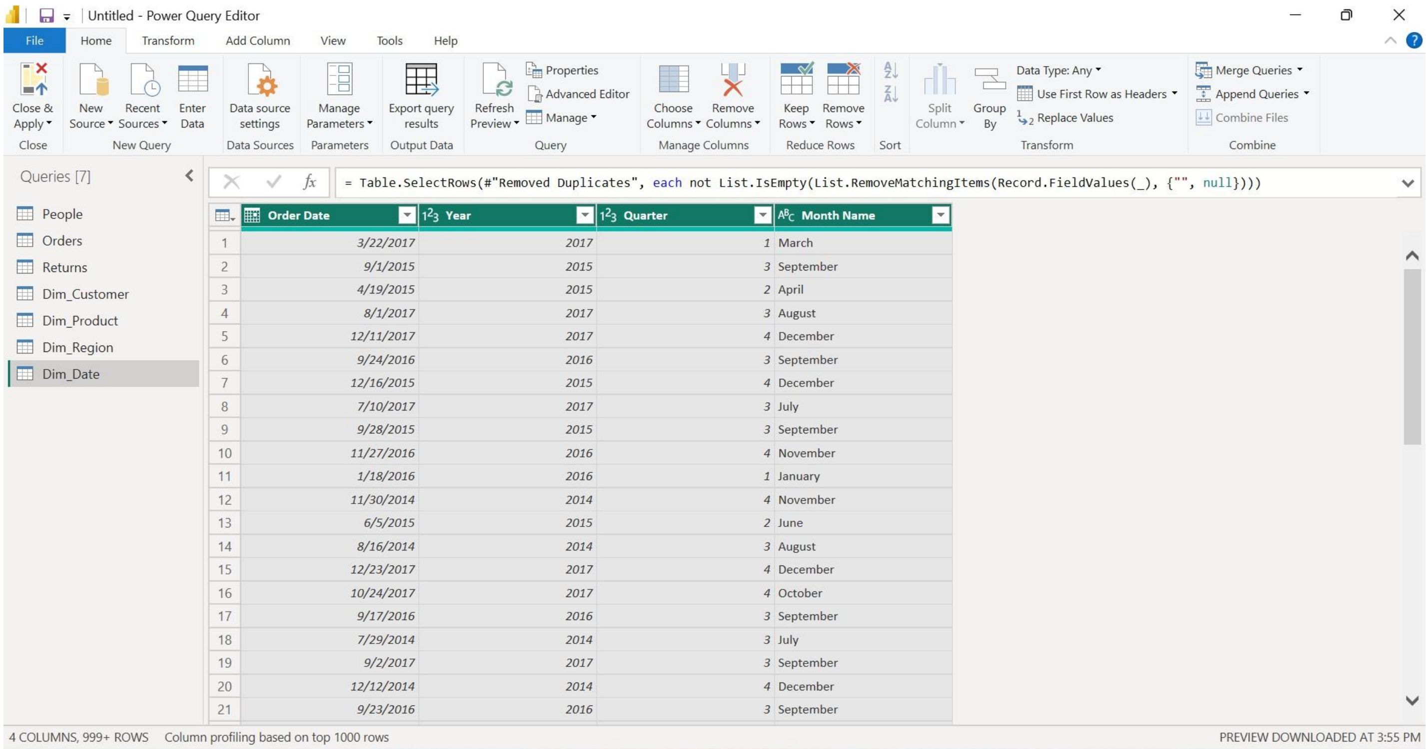
Remove duplicate records.

Step 11: Create Dim_Date.

Create a date dimension containing:

- Order Date
- Year
- Quarter
- Month

Ensure that the date-related fields are correctly formatted.

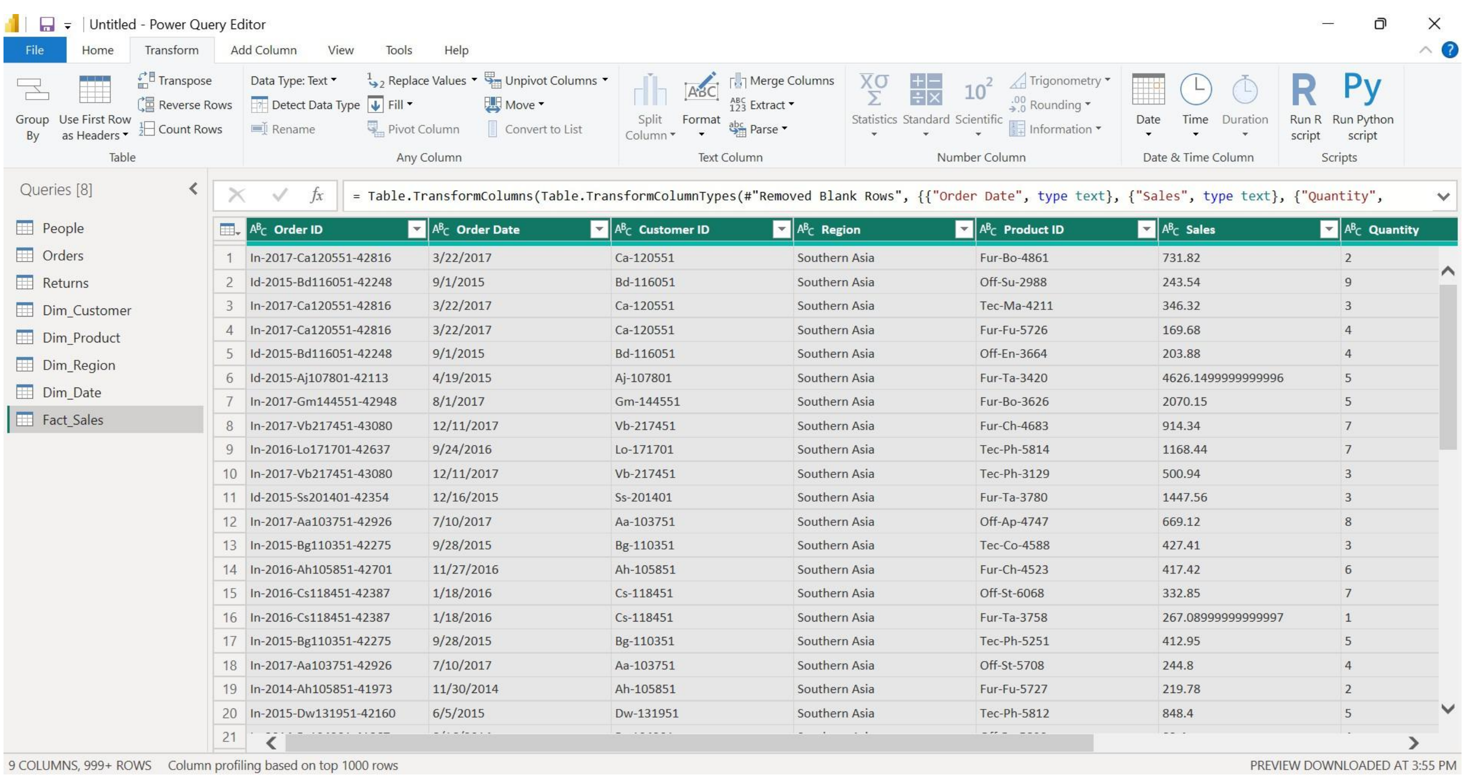


Step 12: Remove duplicate records.

Remove duplicate records from all dimension tables and verify that the key fields contain unique values.

Part D: Create Fact Table

Step 13: Create Fact_Sales.



Create the central Fact Table containing:

- Order ID
- Customer ID
- Product ID
- Region
- Order Date
- Sales
- Quantity
- Profit
- Discount

Step 14: Verify key columns.

Verify that all key columns required to establish relationships with the Dimension Tables are present and correctly formatted.

The resulting Fact_Sales table acts as the central table of the Star Schema.

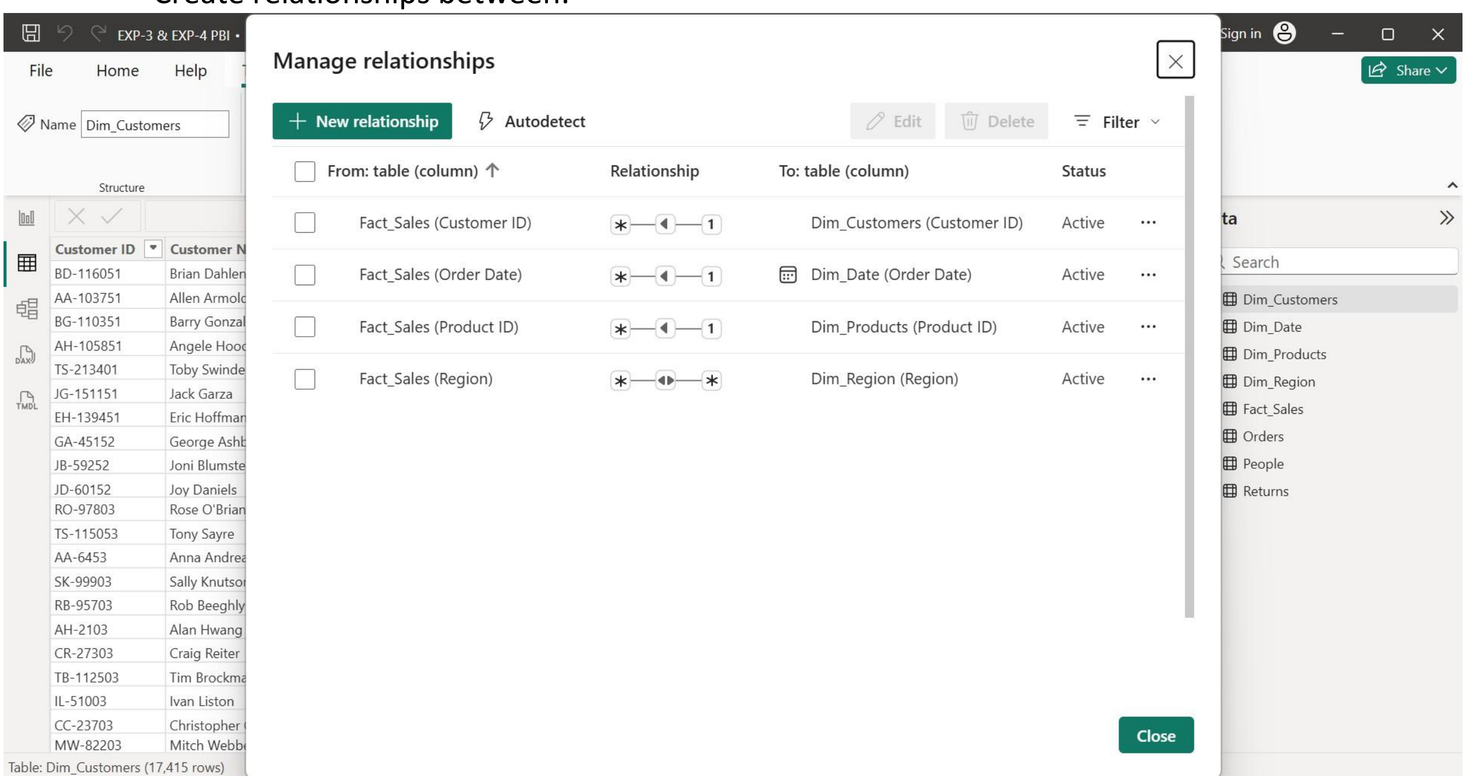
Part E: Build Star Schema

Step 15: Open Model View.

Select the Model View icon from the left navigation pane in Power BI.

Step 16: Create relationships.

Create relationships between:



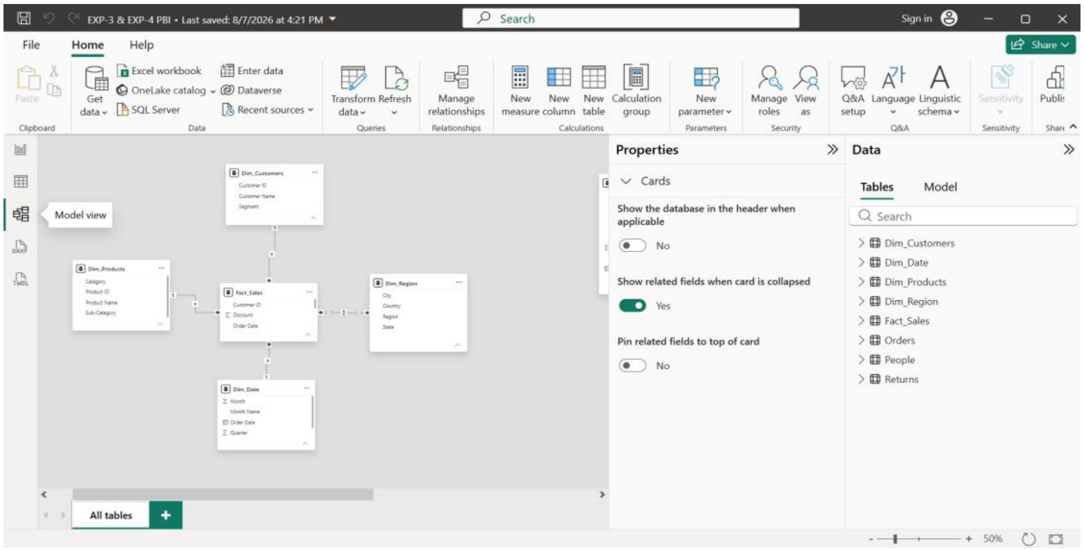
- Fact_Sales ↔ Dim_Customer
- Fact_Sales ↔ Dim_Product
- Fact_Sales ↔ Dim_Region
- Fact_Sales ↔ Dim_Date

Configure relationships.

Set the relationships appropriately as Many-to-One, with the Fact Table on the many side and the corresponding Dimension Table on the one side.

Step 18: Arrange the model.

Arrange the tables visually so that:



Dim_Customer
|
|
Dim_Product ——— Fact_Sales ——— Dim_Region
|
|
Dim_Date

This forms the Star Schema with Fact_Sales at the centre and the Dimension Tables . surrounding it.

Part F: Validate Model and Create DAX Measures

Step 19: Create Power BI visualizations.

Create visualizations using the dimensional model to analyse the data.

Examples include:

- Sales by Category
- Profit by Region
- Sales Trend by Year

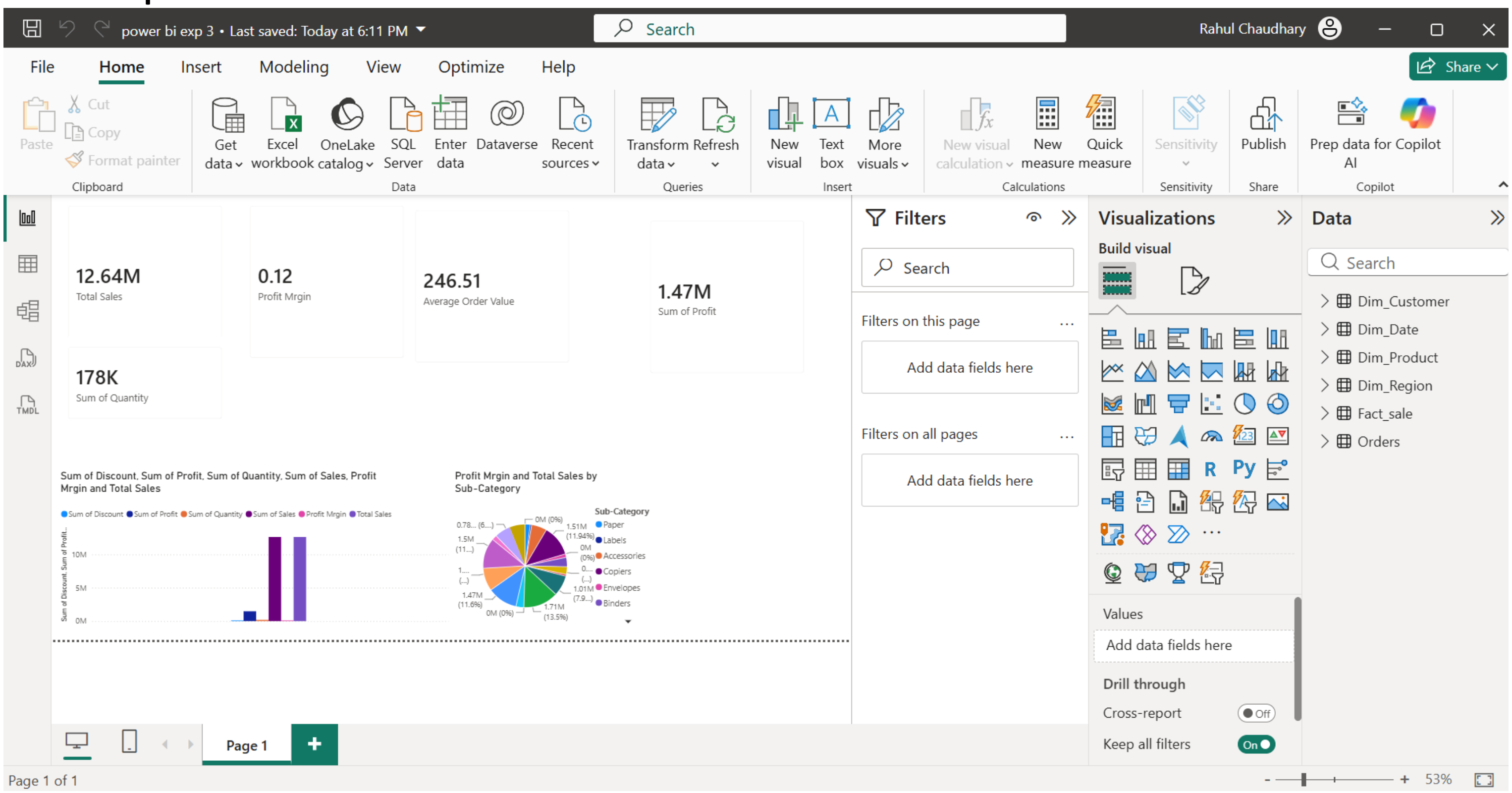
The reference specifies these as examples of visualizations for validating the model.

Step 20: Create DAX measures.

Create the following measures:

1. Total Sales = SUM(Fact_Sales[Sales])
2. Total Profit = SUM(Fact_Sales[Profit])
3. Profit Margin = DIVIDE(SUM(Fact_Sales[Profit]),SUM(Fact_Sales[Sales]))
4. Average Order Value = DIVIDE([Total Sales],DISTINCTCOUNT(Fact_Sales[Order ID]))

Step 21: Validate the Star Schema.



Use the created Dimension Tables to filter the Fact Table and verify that the . .
relationships work correctly.

Check that changes in dimensions such as Customer, Product, Region, and Date correctly affect the corresponding Fact Table data and dashboard visualizations.

6. RESULT

Data was successfully extracted from the Global Superstore dataset, transformed using Power Query Editor, and loaded into the Power BI data model. Data cleaning, standardization, relationship creation, and DAX-based metric generation were performed successfully. The final dataset was prepared for reporting and business analysis.

7. LEARNING OUTCOMES

- Understood the ETL process and its role in Business Intelligence.
- Learned how to extract data from a real-world business dataset using Power BI.
- Gained practical experience with Power Query Editor.
- Learned how to clean, filter, transform, and merge datasets.
- Learned how to handle duplicate records, blank values, and data type inconsistencies.
- Learned how to create relationships between multiple tables in Power BI.
- Learned how to create analytical measures using DAX.
- Understood the use of Total Sales, Total Profit, Profit Margin, and Total Orders measures.
- Learned how to create BI visualizations and dashboards for business analysis.
- Gained practical experience in preparing transformed data for reporting and decision-making.